Project 1

Trishla Nair

2025-09-23

The data we are examining is from CMU's github. We are investigating if sleep patterns can affect GPA of freshman.

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.3
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.4.3
```

```
## corrplot 0.95 loaded
```

```
library(RColorBrewer)

df <- read.csv("C:/Users/Trish/Documents/CMU/CMU-3660/Labs/cmu-sleep.csv")
df <- df%>% select(-subject_id)
print(head(df))
```

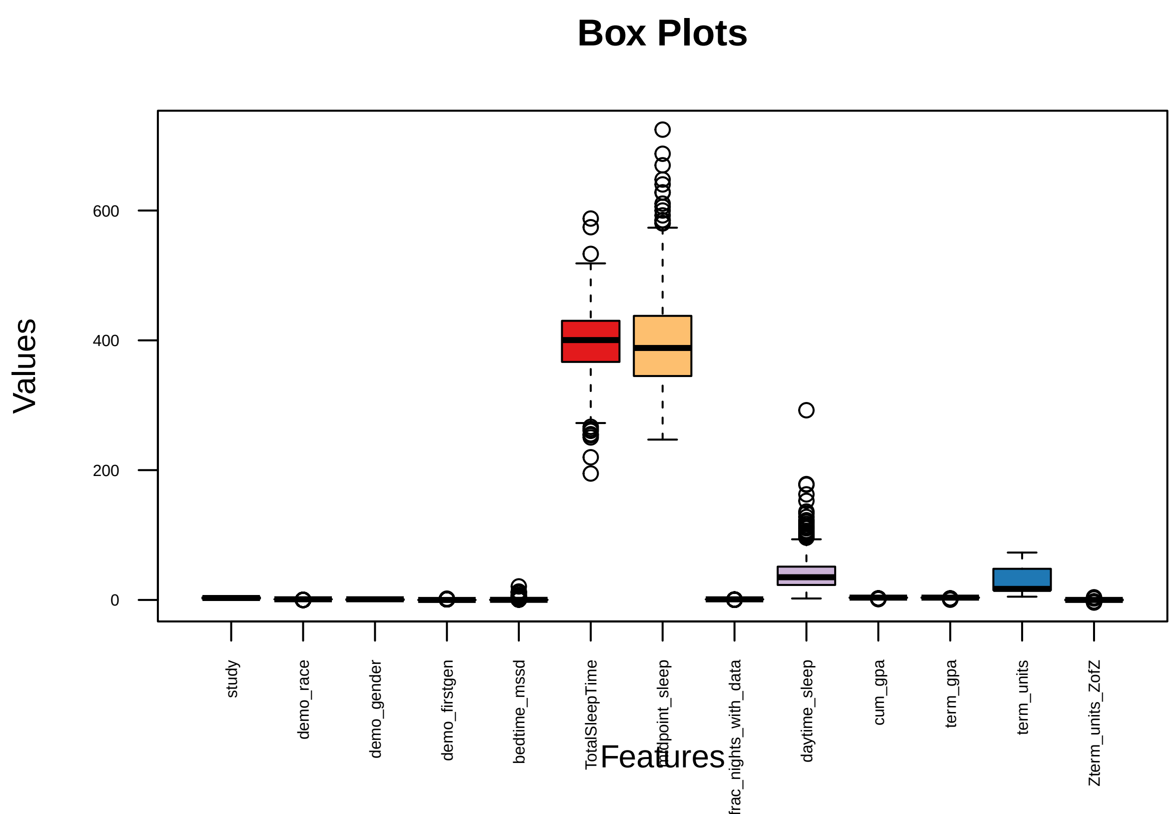
```
##   study cohort demo_race demo_gender demo_firstgen bedtime_mssd TotalSleepTime
## 1     5   lac1         1           1             0    0.1167269       432.2000
## 2     5   lac1         0           1             0    0.1416808       391.9310
## 3     5   lac1         1           1             0    1.5292895       344.3043
## 4     5   lac1         0           1             1    0.1301485       392.6207
## 5     5   lac1         1           1             0    0.1301809       423.4211
## 6     5   lac1         1           1             0    0.2094830       397.4000
```

```
##   midpoint_sleep frac_nights_with_data daytime_sleep cum_gpa term_gpa
## 1       458.6600             0.8620690      24.16000    3.00     3.38
## 2       364.4655             1.0000000      13.13793    3.66     2.60
## 3       560.8913             0.7931034      14.95652    3.57     3.07
## 4       416.4828             1.0000000      54.55172    3.61     3.56
## 5       368.7632             0.6551724      10.52632    3.21     4.00
## 6       353.5800             0.8620690      22.36000    3.20     3.36
##   term_units Zterm_units_ZofZ
## 1         73         4.055295
## 2         64         2.482534
## 3         63         2.307783
## 4         61         1.958281
## 5         61         1.958281
## 6         60         1.783529
```

Firstly, I removed subject id as it had the least relelvance to what I wish to examine (that being the relationship between sleep and cummulative gpa). Then, using only quantitative data, I examined the data for outliers.

```
numeric_cols <- sapply(df, is.numeric)
df_numeric <- df[, numeric_cols]

boxplot(df_numeric, main = "Box Plots",
        xlab = "Features", ylab = "Values", las = 2, cex.axis = 0.5, col = brewer.pal(n = 10, name = "P
```



Based on the boxplots, it seems for the columns related to sleep, have the most outliers. So, next I factorized the demo_ columns and checked for rows with Nan values to remove them.

```r
df <- df %>%
  mutate(cohort = as.factor(cohort),
         demo_gender = as.factor(demo_gender),
         demo_firstgen = as.factor(demo_firstgen),
         demo_race = as.factor(demo_race),
         cohort = as.factor(cohort))
```

Check for NaN's

```r
colSums(sapply(df, is.nan))
```

```
##                study                cohort             demo_race 
##                    0                     0                     0 
##          demo_gender         demo_firstgen          bedtime_mssd 
##                    0                     0                     0 
##       TotalSleepTime        midpoint_sleep frac_nights_with_data 
##                    0                     0                     0 
##        daytime_sleep               cum_gpa              term_gpa 
##                    0                     0                     0 
##           term_units       Zterm_units_ZofZ 
##                    0                     0
```

```r
summary(df)
```

```
##      study        cohort    demo_race  demo_gender demo_firstgen
##  Min.   :1.000   lac1:131   0   :119   0   :263    0   :526    
##  1st Qu.:2.000   lac2: 77   1   :514   1   :368    1   :103    
##  Median :3.000   nh  :147   NA's:  1   NA's:  3    2   :  1    
##  Mean   :3.181   uw1 :140                          NA's:  4    
##  3rd Qu.:4.000   uw2 :139                                      
##  Max.   :5.000                                                 
##                                                                
##   bedtime_mssd       TotalSleepTime  midpoint_sleep  frac_nights_with_data
##  Min.   : 0.004505   Min.   :194.8   Min.   :247.1   Min.   :0.2143       
##  1st Qu.: 0.074694   1st Qu.:366.9   1st Qu.:345.2   1st Qu.:0.8214       
##  Median : 0.135007   Median :400.4   Median :388.2   Median :0.9322       
##  Mean   : 0.451688   Mean   :397.3   Mean   :398.7   Mean   :0.8674       
##  3rd Qu.: 0.291698   3rd Qu.:430.1   3rd Qu.:437.7   3rd Qu.:1.0000       
##  Max.   :20.849225   Max.   :587.7   Max.   :724.7   Max.   :1.0000       
##                                                                           
##  daytime_sleep        cum_gpa         term_gpa       term_units   
##  Min.   :  2.269   Min.   :1.210   Min.   :0.350   Min.   : 5.00  
##  1st Qu.: 23.098   1st Qu.:3.232   1st Qu.:3.233   1st Qu.:15.00  
##  Median : 34.982   Median :3.558   Median :3.556   Median :17.00  
##  Mean   : 41.164   Mean   :3.466   Mean   :3.450   Mean   :29.39  
##  3rd Qu.: 51.249   3rd Qu.:3.790   3rd Qu.:3.810   3rd Qu.:48.00  
##  Max.   :292.304   Max.   :4.000   Max.   :4.000   Max.   :73.00  
##                                                    NA's   :147    
##  Zterm_units_ZofZ  
##  Min.   :-3.98252  
##  1st Qu.:-0.55104  
##  Median : 0.04121
```

```
##  Mean   : 0.00000
##  3rd Qu.: 0.56027
##  Max.   : 4.05529
##  NA's   :147
```

Remove Nan's

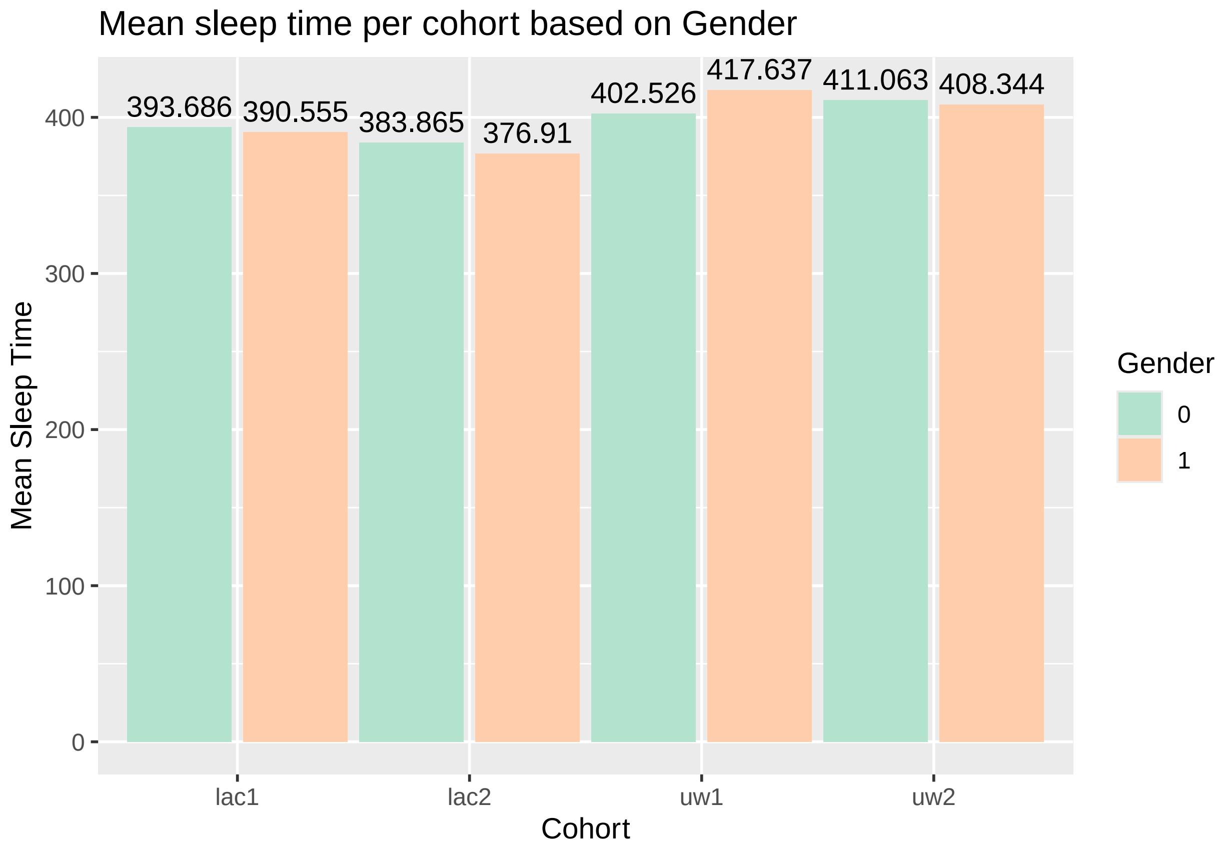
```
df_cleaned <- df %>% drop_na()
summary(df_cleaned)
```

```
##      study       cohort    demo_race demo_gender demo_firstgen
##  Min.   :1.00   lac1:130   0: 88     0:186       0:385
##  1st Qu.:2.00   lac2: 76   1:393     1:295       1: 95
##  Median :3.00   nh  :  0                         2:  1
##  Mean   :2.94   uw1 :137
##  3rd Qu.:5.00   uw2 :138
##  Max.   :5.00
##   bedtime_mssd       TotalSleepTime  midpoint_sleep  frac_nights_with_data
##  Min.   : 0.004505   Min.   :194.8   Min.   :247.1   Min.   :0.2143
##  1st Qu.: 0.075134   1st Qu.:370.5   1st Qu.:345.0   1st Qu.:0.7931
##  Median : 0.142840   Median :402.9   Median :393.1   Median :0.9310
##  Mean   : 0.478967   Mean   :400.9   Mean   :401.9   Mean   :0.8603
##  3rd Qu.: 0.324523   3rd Qu.:433.1   3rd Qu.:442.7   3rd Qu.:1.0000
##  Max.   :20.849225   Max.   :587.7   Max.   :724.7   Max.   :1.0000
##  daytime_sleep        cum_gpa         term_gpa       term_units
##  Min.   :  2.269   Min.   :1.210   Min.   :0.350   Min.   : 5.00
##  1st Qu.: 22.385   1st Qu.:3.160   1st Qu.:3.120   1st Qu.:15.00
##  Median : 34.679   Median :3.500   Median :3.500   Median :17.00
##  Mean   : 40.059   Mean   :3.413   Mean   :3.384   Mean   :29.42
##  3rd Qu.: 50.889   3rd Qu.:3.770   3rd Qu.:3.790   3rd Qu.:48.00
##  Max.   :292.304   Max.   :4.000   Max.   :4.000   Max.   :73.00
##  Zterm_units_ZofZ
##  Min.   :-3.982521
##  1st Qu.:-0.604268
##  Median : 0.041207
##  Mean   :-0.000448
##  3rd Qu.: 0.560271
##  Max.   : 4.055295
```

I then built a bar graph of mean sleep of each cohort per gender to see if there possibly wether the school or gender may yeild interesting insights on the average sleep a freshamn gets.

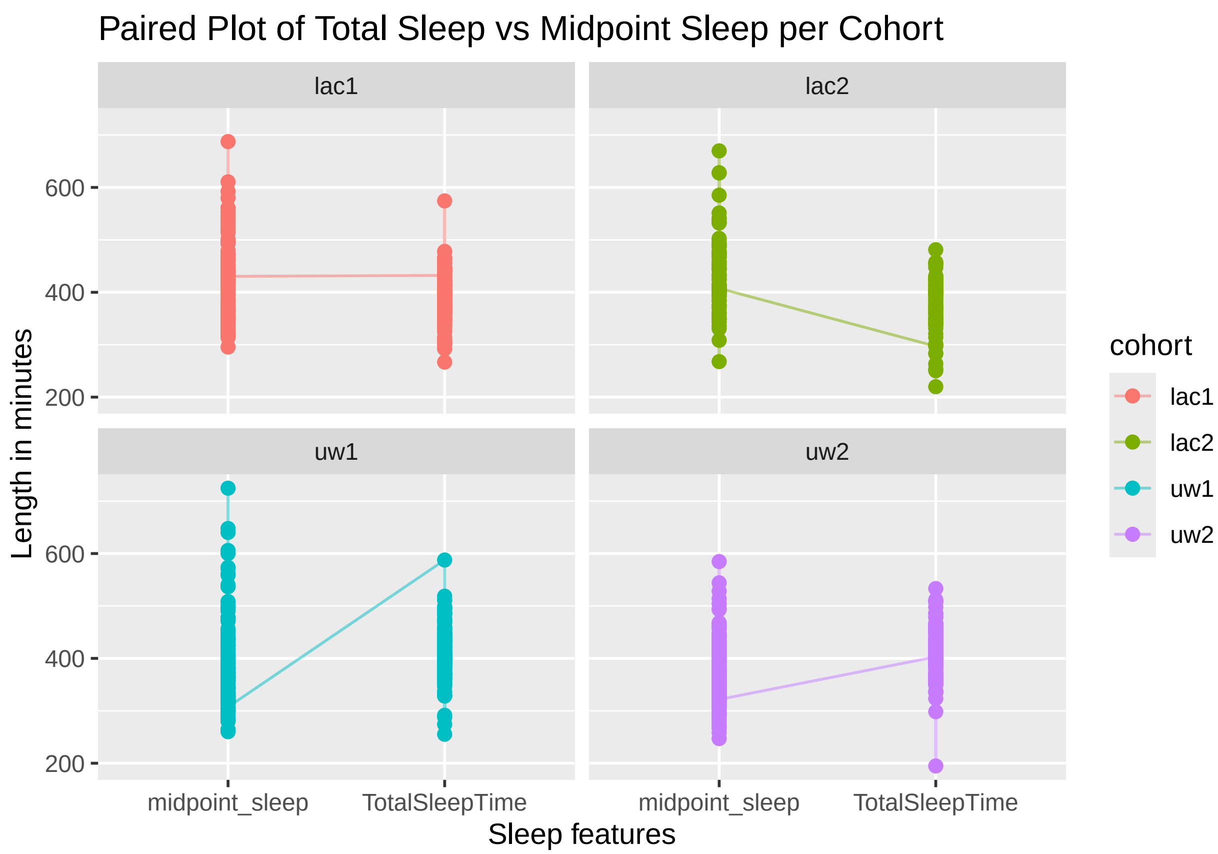
```
#
pl<- df_cleaned %>%group_by(cohort, demo_gender) %>%
  summarize(mean_sleep = mean(TotalSleepTime, na.rm = TRUE), .groups = 'drop')

ggplot(pl, aes(x = cohort, y = mean_sleep, fill = demo_gender)) +
  geom_col(position = position_dodge(width = 1)) +
  geom_text(aes(label = round(mean_sleep, 3)),
            position = position_dodge(width = 1),
            vjust = -0.5) +
  scale_fill_brewer(palette = "Pastel2") +
  labs(title = "Mean sleep time per cohort based on Gender",x = "Cohort", y = "Mean Sleep Time", fill =
```



I then constructed a paired plot examining total sleep and midpoint sleep per cohort to see if the school plays a factor into wether sleep affects gpa.

```
sleep <- df_cleaned %>%
  pivot_longer(cols = c(TotalSleepTime, midpoint_sleep),
               names_to = "Measure",
               values_to = "Value")
ggplot(sleep, aes(x = Measure, y = Value, group = cohort, color = cohort)) +
  geom_line(alpha = 0.5) +
  geom_point(size = 2) +
  facet_wrap(~ cohort) +   # <-- split into panels
  labs(title = "Paired Plot of Total Sleep vs Midpoint Sleep per Cohort",
       x = "Sleep features",
       y = "Length in minutes")
```



Based on the pair plots, the UW cohorts seem to sleep longer compared to the LAC cohorts, as indicated by the fact the LAC cohorts seems to have midpoint_sleep time much later than the UW students. I then did a chi-square test to examine the possible correlation between mean sleep time and mean gpa.

```
cor.test(df_cleaned$TotalSleepTime, df_cleaned$term_gpa, use = "complete.obs")
```

```
##
##  Pearson's product-moment correlation
##
## data:  df_cleaned$TotalSleepTime and df_cleaned$term_gpa
## t = 5.5371, df = 479, p-value = 5.077e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.1593551 0.3274937
## sample estimates:
##       cor
## 0.2452679
```

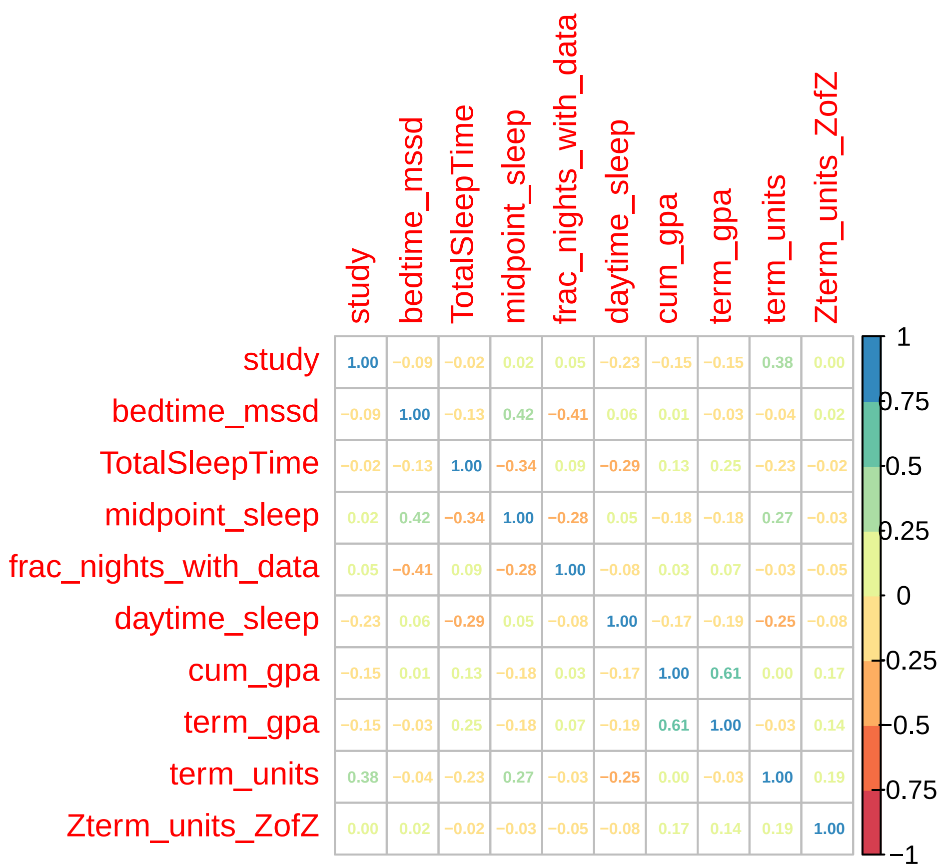
```
lm_gpa <- lm(term_gpa ~ TotalSleepTime, data = df_cleaned)
summary(lm_gpa)
```

```
##
## Call:
## lm(formula = term_gpa ~ TotalSleepTime, data = df_cleaned)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8626 -0.2433  0.1087  0.4038  0.8874
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.3372831  0.1905405  12.267  < 2e-16 ***
## TotalSleepTime 0.0026110  0.0004715   5.537 5.08e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5219 on 479 degrees of freedom
## Multiple R-squared:  0.06016,	Adjusted R-squared:  0.05819
## F-statistic: 30.66 on 1 and 479 DF,  p-value: 5.077e-08
```

Based on my findings, since the p value is so small, we reject the null hypothesis and see that there is a significant correlation between the 2. Thus, I created a correlation matrix to see what other variables might affect term_gpa.

```r
num_cols <- sapply(df_cleaned, is.numeric)
df_num <- df_cleaned[, num_cols]
matrix<-cor(df_num)
corrplot(matrix, method="number", number.cex = 0.5, col = brewer.pal(n = 8, name = "Spectral"))
```



After running a correlation matrix, the only feature that may possess a strong correlation with term GPA is the cumulative GPA. I then ran a hypothesis test to examine whether the time may effect the term gpa by comparing the first and fifth studies at a 95% confidence level.

```
s_1 <- df_cleaned%>%select(term_gpa, study)%>%filter(study == 1)
s_5 <- df_cleaned%>%select(term_gpa, study)%>%filter(study == 5)
t_test_result <- t.test(s_1$term_gpa, s_5$term_gpa,alternative = "less", var.equal = FALSE, conf.level =

print(t_test_result)
```

```
##
##  Welch Two Sample t-test
##
## data:  s_1$term_gpa and s_5$term_gpa
## t = 2.5779, df = 171.94, p-value = 0.9946
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##       -Inf 0.3285446
## sample estimates:
## mean of x mean of y
##  3.477763  3.277615
```

Given how large the pvalue is, we fail to reject the null hypothesis and thus there is no difference in term GPA's between studies 1 and 5.